Q9 $I = 2.2\,A \quad B = 14.0\,T \quad \theta = 37^\circ \quad L = 0.02\,m$

$F_m = ?$

$F_m = I L B \sin\theta = 2.2 \times 0.02 \times 14 \times \sin 37$

$= 0.371\,N$

Q10 $l = 96\,m \quad d = 0.0019\,m \quad I = 233\,A$

$F = ?$

$\mu_{air} = 4\pi \times 10^{-7}$

$F = \dfrac{\mu I_1 I_2 l}{2\pi d} = \dfrac{4\pi \times 10^{-7} \times 233 \times 233 \times 96}{2 \times 0.0019}$

$= 5480.605\,N$